

MIT OpenCourseWare

18.06SC

Linear Algebra 18.06SC — Linear Algebra May 2026 **EXTREMELY HARD – MIT****LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (4 points) Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be a sequence of functions of bounded variation with $\text{Var}(f_n) \leq C$ uniformly and $f_n \rightarrow f$ pointwise. Prove that f is of bounded variation and $\text{Var}(f) \leq C$. Give a counterexample showing that uniform bounded variation is necessary.
2. (2 points) Prove that there is no continuous linear functional $L : \ell^\infty \rightarrow \mathbb{R}$ that extends the limit functional on c (convergent sequences) and satisfies $\|L\| = 1$. (Hint: use the Hahn-Banach theorem and consider the Banach limit.)
3. (5 points) Classify the second-order linear PDE $u_{xx} + 2u_{xy} + u_{yy} + u_x + u_y = 0$. Determine its type (elliptic, parabolic, hyperbolic) and find its canonical form. Solve the resulting canonical equation.
4. (6 points) Prove the Nash embedding theorem for compact Riemannian manifolds: any compact C^∞ Riemannian manifold can be isometrically embedded into $\mathbb{R}^N$ for sufficiently large N .
5. (5 points) Prove the Hille-Yosida theorem: a linear operator A generates a strongly continuous contraction semigroup on a Banach space X iff A is closed, densely defined, and all $\lambda > 0$ are in the resolvent set with $\|R(\lambda, A)\| \leq 1/\lambda$.
6. (2 points) Prove the Rellich-Kondrachov compactness theorem: $W^{1,p}(\Omega) \hookrightarrow L^q(\Omega)$ compactly for $1 \leq q < p^*$ when Ω is bounded and $\partial\Omega$ is C^1 .
7. (4 points) Prove Sard's theorem: if $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$ is C^k with $k > \max(n - m, 0)$, then the set of critical values of f has Lebesgue measure zero in $\mathbb{R}^m$.
8. (3 points) Prove the Poincaré inequality: for $u \in W_0^{1,p}(\Omega)$ with $\Omega \subset \mathbb{R}^n$ bounded, $\|u\|_{L^p} \leq C \|\nabla u\|_{L^p}$. Show that the constant C depends on the diameter of Ω and give the optimal constant for $p = 2$ on an interval.
9. (2 points) Prove the Brouwer fixed-point theorem using differential topology: any continuous map $f : \overline{B}^n \rightarrow \overline{B}^n$ has a fixed point. Use the fact that there is no smooth retraction of the ball onto its boundary.

10. (2 points) Prove that the Fourier transform is an isometry on $L^2(\mathbb{R}^n)$ (Plancherel's theorem) and extends uniquely from $L^1 \cap L^2$ to a unitary operator on L^2 .
11. (6 points) Determine all values of $p \in \mathbb{R}$ for which the series $\sum_{n=2}^{\infty} \frac{1}{n^p \ln n}$ converges. Prove your answer using the integral test and comparison tests.
12. (7 points) State and prove the spectral theorem for compact self-adjoint operators on a Hilbert space. Show that such an operator has a countable set of eigenvalues that accumulate only at zero, and the eigenvectors form an orthonormal basis.
13. (2 points) Prove that symplectic integrators preserve the symplectic structure $\omega = \sum dq^i \wedge dp_i$ of Hamiltonian systems. Show that the Verlet (Störmer) method is symplectic and has energy error bounded over exponentially long times for integrable systems.
14. (2 points) Prove the Lax-Milgram theorem and use it to show existence and uniqueness of weak solutions to the Dirichlet problem $-\Delta u = f$ in Ω , $u|_{\partial\Omega} = 0$ for $f \in L^2(\Omega)$.
15. (4 points) Let M be a compact oriented n -manifold with boundary ∂M . State and prove Stokes' theorem for differential forms. Then apply it to prove the divergence theorem: $\int_M \operatorname{div} F \, dV = \int_{\partial M} F \cdot n \, dS$.
16. (2 points) Prove the Hopf-Rinow theorem: for a Riemannian manifold (M, g) , metric completeness is equivalent to geodesic completeness (every geodesic extends for all time).
17. (4 points) Prove the open mapping theorem: if $T : X \rightarrow Y$ is a bounded linear operator between Banach spaces and T is surjective, then T maps open sets to open sets. Use it to prove the inverse mapping theorem.
18. (5 points) Construct an exhaustion function (proper smooth function bounded below) on any smooth manifold and use it to prove that every smooth manifold admits a complete Riemannian metric.
19. (5 points) Prove that the Laplace operator Δ on a compact Riemannian manifold has discrete spectrum $\lambda_1 \leq \lambda_2 \leq \dots \rightarrow \infty$ and each eigenvalue has finite multiplicity. Show $\lambda_1 = \inf \frac{\|\nabla u\|^2}{\|u\|^2}$ over nonzero $u \in H^1(M)$ with $\int u = 0$.
20. (3 points) Prove the Divergence theorem for smooth vector fields on compact manifolds with boundary and derive the Green's identities.
21. (4 points) Compute the Fourier transform of the distribution $p.v.(1/x)$ and prove that $\widehat{p.v.(1/x)}(\xi) = -i\pi \operatorname{sgn}(\xi)$. Use this to solve the Hilbert transform equation $H(u) = f$.

22. (3 points) Prove the Gagliardo-Nirenberg-Sobolev inequality: for $1 \leq p < n$, there exists $C = C(n, p) > 0$ such that $\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C \|\nabla u\|_{L^p(\mathbb{R}^n)}$ for all $u \in C_c^1(\mathbb{R}^n)$, where $p^* = np/(n - p)$.
23. (4 points) State and prove the Implicit Function Theorem. Apply it to show that the equation $F(x, y) = y^5 + y + x = 0$ can be solved for $y = g(x)$ near $(0, 0)$, and compute $g'(0)$.
24. (8 points) Prove the maximum principle for elliptic PDEs: if $Lu = -\Delta u + c(x)u \geq 0$ in Ω with $c \geq 0$, then $\max_{\overline{\Omega}} u \leq \max_{\partial\Omega} u^+$. Show how this implies uniqueness for the Dirichlet problem.
25. (6 points) Construct a sequence of continuous functions $f_n : [0, 1] \rightarrow \mathbb{R}$ that converges pointwise to $f(x) = 0$ but does NOT converge uniformly. Prove your construction works and explain why uniform convergence fails.

Solutions

Solution 1:

For any partition $0 = x_0 < \dots < x_k = 1$, $|f(x_i) - f(x_{i-1})| = \lim_n |f_n(x_i) - f_n(x_{i-1})| \leq \liminf_n \text{Var}(f_n) \leq C$. So $\text{Var}(f) \leq C$. Counterexample without UB: $f_n(x) = nx$ on $[0, 1/n]$, $f_n(x) = 1$ elsewhere. $\text{Var}(f_n) = 2n$, $f_n \rightarrow 1$ pointwise, but limit has variation 0, so inequality fails direction.

Solution 2:

The Hahn-Banach theorem guarantees the existence of such extensions (Banach limits), but they are not unique and not continuous in the sense of being representable by a sequence. The key issue: ℓ^∞/c_0 is non-separable and the quotient map is not injective. Banach limits exist but require the axiom of choice. They satisfy $\liminf x_n \leq L(x) \leq \limsup x_n$ and are shift-invariant.

Solution 3:

The characteristic equation: $a_{11}dy^2 - 2a_{12}dxdy + a_{22}dx^2 = 0$ gives $1 \cdot dy^2 - 2 \cdot 1 \cdot dxdy + 1 \cdot dx^2 = 0$, so $(dy - dx)^2 = 0$. Double root: parabolic type. Characteristic curves: $y - x = c$. Canonical coordinates: $\xi = y - x$, $\eta = x$. Then $u_{\eta\eta} + u_\eta = 0$. Solve: $u(\xi, \eta) = A(\xi)e^{-\eta} + B(\xi)$, i.e. $u(x, y) = A(y - x)e^{-x} + B(y - x)$.

Solution 4:

Sketch: Use the implicit function theorem to construct a short immersion $f_0 : M \rightarrow \mathbb{R}^N$. The metric g on M can be expressed as $f_0^*g_{\mathbb{R}^N} + h$. Use the Nash-Moser iteration to solve the system of PDEs $\partial_i f \cdot \partial_j f = g_{ij}$. The iteration converges by tame estimates on the linearized operator $L_f(h) = df \cdot \nabla h$.

Solution 5:

($\Rightarrow$): If A generates $T(t)$, show $R(\lambda, A) = \int_0^\infty e^{-\lambda t} T(t) dt$, so $\|R(\lambda, A)\| \leq \int_0^\infty e^{-\lambda t} \|T(t)\| dt \leq 1/\lambda$. ($\Leftarrow$): Define $A_n = nAR(n, A)$ (Yosida approximants) which are bounded generators of $T_n(t) = e^{tA_n}$. Show $T_n(t)$ converges strongly to a semigroup $T(t)$ and its generator is A .

Solution 6:

Take a bounded sequence $\{u_n\}$ in $W^{1,p}$. By the Sobolev-Gagliardo-Nirenberg inequality, it's bounded in L^{p^*} . Use the Frechet-Kolmogorov criterion for compactness in L^q : show $\|u_n(\cdot + h) - u_n\|_{L^q} \rightarrow 0$ uniformly as $h \rightarrow 0$ using difference quotients and the $W^{1,p}$ bound. Extract a subsequence convergent in L^q by the Arzela-Ascoli type argument.

Solution 7:

Cover U by cubes. On each cube where $\|Df\| \leq C$ and $|\det Df| \leq \varepsilon$ on a critical point, use the Whitney extension to show the image can be covered by $O((\varepsilon/C)^m)$ balls of radius $C/\sqrt{n}$. Summing over cubes gives arbitrarily small total measure. The differentiability condition ensures the critical set has enough structure.

Solution 8:

For $u \in C_c^\infty(\Omega)$, write $u(x) = \int_{-\infty}^{x_1} \partial_1 u(t, x'') dt$. By Hölder, $|u(x)|^p \leq (\text{diam } \Omega)^{p-1} \int |\partial_1 u|^p dt$. Integrate over Ω to get $\|u\|_p^p \leq (\text{diam } \Omega)^p \|\nabla u\|_p^p$. On $(0, \pi)$, the optimal constant for $p = 2$

is $C = 1/\pi$, attained by $u(x) = \sin(x)$.

Solution 9:

Suppose f has no fixed point. Define $r(x)$ = intersection of ray from $f(x)$ through x with ∂B^n . This gives a smooth retraction (after smooth approximation). By Sard's theorem, regular values exist. For a regular value y , $r^{-1}(y)$ is a 1-manifold with boundary, one endpoint at y . The other endpoint would be in B^n , impossible since r maps interior to boundary. Contradiction.

Solution 10:

For $f \in L^1 \cap L^2$, define $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x \cdot \xi} dx$. Show $\int |\widehat{f}|^2 = \int |f|^2$ by proving $\int \widehat{f}\widehat{g} = \int f\bar{g}$ for g Schwartz (using Gaussian approximation and Fubini). Extend by density of $L^1 \cap L^2$ in L^2 . The inverse Fourier transform gives the adjoint, and $\mathcal{F}^* = \mathcal{F}^{-1}$.

Solution 11:

By the integral test, $\int_2^\infty \frac{dx}{x^p \ln x}$ converges iff $p > 1$. For $p = 1$, $\int_2^\infty \frac{dx}{x \ln x} = \lim_{b \rightarrow \infty} \ln(\ln b) - \ln(\ln 2) = \infty$, so diverges. For $p < 1$, compare with $1/x^p$: $\frac{1}{n^p \ln n} > \frac{1}{n^p \cdot n^\varepsilon}$ for large n , which diverges by p-test. For $p > 1$, $\frac{1}{n^p \ln n} < \frac{1}{n^{(p+1)/2}}$ for large n , which converges.

Solution 12:

Let $T : H \rightarrow H$ be compact self-adjoint. Show $\|T\| = \sup_{|x|=1} |\langle Tx, x \rangle|$ (achieved). Let $\lambda_1 = \pm\|T\|$ be the extremal eigenvalue with eigenvector e_1 . Inductively apply the same argument to T restricted to $\{e_1, \dots, e_{n-1}\}^\perp$. Compactness implies $\lambda_n \rightarrow 0$. The eigenvectors form an ONB by the spectral decomposition $Tx = \sum \lambda_n \langle x, e_n \rangle e_n$.

Solution 13:

A one-step method Φ_h is symplectic if $(\Phi_h)^*\omega = \omega$. For Verlet: $q_{n+1/2} = q_n + \frac{h}{2}M^{-1}p_n$, $p_{n+1} = p_n - h\nabla V(q_{n+1/2})$, $q_{n+1} = q_{n+1/2} + \frac{h}{2}M^{-1}p_{n+1}$. Direct computation shows the Jacobian satisfies $J^T \Omega J = \Omega$. By the backward error analysis, the numerical solution is the exact flow of a nearby Hamiltonian, giving $O(e^{-c/h})$ energy conservation over exponentially long times.

Solution 14:

Lax-Milgram: Let $a : H \times H \rightarrow \mathbb{R}$ be coercive bilinear continuous on Hilbert H , $f \in H^*$. Then $\exists! u \in H$: $a(u, v) = f(v)$ for all $v \in H$. For the Dirichlet problem, $H = H_0^1(\Omega)$, $a(u, v) = \int_\Omega \nabla u \cdot \nabla v$, $f(v) = \int_\Omega f v$. Poincaré inequality gives coercivity, trace theorem gives boundary values.

Solution 15:

Stokes: $\int_M d\omega = \int_{\partial M} \omega$ for an $(n-1)$ -form ω . Proof: Use partition of unity to reduce to $\mathbb{R}^n$ case, where it follows from the fundamental theorem of calculus and Fubini. For divergence theorem, let $\omega = \sum_i (-1)^{i-1} F_i dx^1 \wedge \dots \wedge \widehat{dx^i} \wedge \dots \wedge dx^n$. Then $d\omega = (\sum_i \partial F_i / \partial x^i) dx^1 \wedge \dots \wedge dx^n = \operatorname{div} F dV$. The boundary term gives $\int_{\partial M} F \cdot n dS$ by the Hodge star duality.

Solution 16:

$(\Rightarrow)$: Assume metric completeness. For a geodesic γ with maximal interval (a, b) , if $b < \infty$, take $t_n \rightarrow b$. Then $\{\gamma(t_n)\}$ is Cauchy (distance $\leq C|t_n - t_m|$) so converges. Extend

γ using the exponential map at the limit point. ($\Leftarrow$): Assume geodesic completeness. Take a Cauchy sequence p_n . Connect each p_n to p_0 by a minimizing geodesic. Bound on distances and completeness of initial data gives convergent subsequence.

Solution 17:

Use Baire category: $Y = \cup_n nT(B_X(0, 1))$ so some $nT(B_X(0, 1))$ has nonempty interior. Show $T(B_X(0, 1))$ contains a neighborhood of 0. Then by scaling, T maps open balls to open sets. The inverse mapping theorem follows: if T is bijective bounded linear, then T^{-1} is bounded.

Solution 18:

Let $\{U_i\}$ be a countable cover by coordinate balls, $\{\rho_i\}$ a partition of unity. Define $f = \sum_i \rho_i$. This is proper: $f^{-1}([-\infty, c])$ is compact (only finitely many $i \leq c$ contribute). For a Riemannian metric, start with any metric g_0 , let $g = g_0 + df \otimes df$ on $f^{-1}((-\infty, \infty))$. The gradient of f has unit length, so any curve must escape to infinity in infinite length.

Solution 19:

The resolvent $(-\Delta + I)^{-1} : L^2 \rightarrow H^2$ is compact by Rellich-Kondrachov. The spectral theorem for compact self-adjoint operators gives discrete eigenvalues $\mu_n \rightarrow 0$, so $\lambda_n = 1/\mu_n - 1 \rightarrow \infty$. Finite multiplicity follows from finite-dimensional eigenspaces of compact operators. The variational characterization follows from the Rayleigh quotient. The condition $\int u = 0$ removes the trivial $\lambda_0 = 0$.

Solution 20:

For $F \in \mathfrak{X}(\Omega)$, choose a partition of unity subordinated to a cover by coordinate charts that are half-balls near boundary. In each chart, the divergence theorem reduces to integrating a partial derivative over a cube (or half-cube), which by Fubini and FTC gives boundary integrals. Summing yields $\int_\Omega \operatorname{div} F \, dV = \int_{\partial\Omega} F \cdot n \, dS$. Green's identities follow by setting $F = u \nabla v$.

Solution 21:

$\langle p.v.(1/x), \varphi \rangle = \langle p.v.(1/x), \widehat{\varphi} \rangle$. Write $p.v.(1/x) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{x} \chi_{|x| > \varepsilon}$. Its Fourier transform: $\int_{|x| > \varepsilon} \frac{e^{-2\pi i x \xi}}{x} dx = -i\pi \operatorname{sgn}(\xi)$ as $\varepsilon \rightarrow 0$. The Hilbert transform $H(u) = \frac{1}{\pi} p.v. \int \frac{u(t)}{x-t} dt$ has symbol $\widehat{H(u)}(\xi) = -i \operatorname{sgn}(\xi) \widehat{u}(\xi)$, so $H^{-1} = -H$.

Solution 22:

For $p = 1$, use $|u(x)| \leq \int_{-\infty}^{x_i} |\partial_i u| dx_i$, multiply over $i = 1, \dots, n$, integrate, apply Hölder repeatedly to get $\|u\|_{n/(n-1)} \leq \prod (\int |\partial_i u|)^{1/n} \leq \frac{1}{n} \sum \int |\partial_i u|$. For general p , apply the $p = 1$ case to $|u|^q$ with suitable q and use Hölder.

Solution 23:

IFT: Let $F : \mathbb{R}^{m+n} \rightarrow \mathbb{R}^n$ be C^1 , $F(a, b) = 0$, and $\partial F / \partial y(a, b)$ invertible. Then there exists $g : U \rightarrow \mathbb{R}^n$ C^1 near a with $F(x, g(x)) = 0$ and $Dg(x) = -(\partial F / \partial y)^{-1} \partial F / \partial x$. For $F(x, y) = y^5 + y + x$, $\partial F / \partial y = 5y^4 + 1$, at $(0, 0)$ this is $1 \neq 0$, so IFT applies. $g'(0) = -(\partial F / \partial y)^{-1} \partial F / \partial x = -1/1 = -1$.

Solution 24:

Suppose u attains an interior maximum at x_0 with $u(x_0) = M > 0$ and $M > \max_{\partial\Omega} u^+$. At x_0 , $\Delta u \leq 0$ (Hessian negative semidefinite), so $Lu = -\Delta u + cu \geq 0 + cM \geq 0$. Strictly, if $c > 0$ or $M > 0$, this gives $Lu > 0$ at the maximum, contradiction. For uniqueness of $\Delta u = f$ with $u|_{\partial\Omega} = g$, difference of two solutions satisfies $\Delta w = 0$, $w|_{\partial\Omega} = 0$, so $w \equiv 0$.

Solution 25:

Let $f_n(x) = \max(0, 1 - n|x - 1/n|)$ (triangular spikes at $1/n$). For any x , for $n > 1/x$, $f_n(x) = 0$, so $f_n \rightarrow 0$ pointwise. But $\|f_n\|_\infty = 1$ for all n , so $\sup_{x \in [0,1]} |f_n(x) - 0| = 1 \not\rightarrow 0$, hence convergence is not uniform. This shows pointwise convergence does not imply uniform convergence.